4.10 Differential Equations				
4.10a	General and particular solutions		a) Understand the difference between, and be able to find, general and particular solutions to differential equations. *Includes understanding that the general solution will include arbitrary constant(s) and that the particular solution may be found from initial or boundary conditions.*	I2
4.10b	Modelling		b) Be able to use differential equations in modelling in kinematics and in other contexts. *Includes use of Newton's second law of motion and the language of kinematics including the notation* $v = \dot{x} = \frac{\mathrm{d}x}{\mathrm{d}t}$ *and* $a = \dot{v} = \ddot{x} = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$. *Includes problems involving variable force.* *Problems may include formulating differential equations which leaners cannot solve analytically.* [*Problems involving either variable mass or the form* $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$ *are excluded.*]	I3, I2
4.10c	Integrating factor method for first order differential equations		c) Be able to find and use an integrating factor $\mathrm{e}^{\int \mathrm{P}(x)\mathrm{d}x}$ to solve differential equations of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$. *Includes recognising when it is appropriate to do so, rearranging into the given form when necessary.*	I1

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.10d	Second order homogeneous differential equations		d) Be able to solve differential equations of the form $y'' + ay' + by = 0$, where a and b are constants, by using the auxiliary equation. *Includes rearranging into the given form when necessary.* *Learners should be able to interpret the sign of the discriminant of the auxiliary equation and how it determines the form of the complementary function.* *Including the cases when the roots of the auxiliary equation are:* *(i) distinct and real,* *(ii) repeated,* *(iii) complex.*	I4 I6 I2
4.10e	Second order non-homogeneous differential equations		e) Be able to solve differential equations of the form $y'' + ay' + by = \mathrm{f}(x)$, where a and b are constants, by solving the homogeneous case and adding a particular integral to the complementary function (in cases where $\mathrm{f}(x)$ is a polynomial, exponential or trigonometric function). *Includes cases where the form of the complementary function affects the choice of trial integral for the particular integral.* *Includes cases where the form of the particular integral is given.*	I5

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…	DfE Ref.
4.10f	Simple harmonic motion		f) Be able to solve the equation for simple harmonic motion (SHM) $\ddot{x} = -\omega^2 x$ and relate the solution to the motion. *Includes use of the formulae* $x = A\cos(\omega t) + B\sin(\omega t)$ *and* $x = R\sin(\omega t + \varphi)$ *in* *modelling situations.* *Learners may quote these formulae without proof when not asked to derive it or to solve the SHM equation.*	I7
4.10g	Damped oscillations		g) Be able to model damped oscillations using second order differential equations and interpret their solutions. *The terms "underdamping", "overdamping" and "critical damping" should be known and understood informally.*	I8
4.10h	Linear systems		h) Be able to analyse and interpret models of situations with one independent variable and two dependent variables as a pair of coupled, simultaneous, first order differential equations, and be able to solve them. *e.g. Predator-prey models, continuous population models, industrial processes.* *Includes solution by eliminating one variable to produce a single second order differential equation.* *Systems will be of the form* $\frac{\mathrm{d}x}{\mathrm{d}t} = ax + by + \mathrm{f}(t)$, $\frac{\mathrm{d}y}{\mathrm{d}t} = cx + dy + \mathrm{g}(t)$ *or easily reducible to this form.*	I9